

DOCeV: A New Holistic Model for Direct Operating Cost Estimation of Electric Vertical Take-Off and Landing Aircraft[☆]

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ABSTRACT

Technological advancements foster the development of novel aircraft concepts such as electric vertical take-off and landing aircraft (eVTOLs). As a result of their promised operational benefits, such as low-emission operations and travel time savings, eVTOLs have the potential to complement or substitute existing ground transportation modes. However, their market penetration predominantly depends on economically viable operation. A widely used indicator for evaluating economic performance in aviation is the Direct Operating Cost (DOC). Currently, there is only limited public documentation on methodologies for estimating DOC of eVTOLs, and existing approaches are often abstract. Therefore, this article introduces the DOCeV, a model for estimating eVTOL DOC. The model is based on a reference methodology for conventional aircraft but is modified and extended to address the specific characteristics of (future) eVTOL operations. The DOCeV enables the estimation of costs associated with ownership, insurance, crew, energy, fees, and maintenance. Additionally, an eVTOL performance analysis is conducted within the model, the annual eVTOL utilisation is estimated, and an indicative ticket pricing approach is provided. The DOCeV has been implemented in a tool and validated using an input data set developed for a regional airport shuttle case study between Munich, Germany, and Salzburg, Austria. When applying the DOCeV, the DOC estimates for shorter, regional operations align well with existing literature. Estimates for longer missions are slightly higher than those reported in the literature. Overall, the DOCeV is considered a comprehensive and promising model for estimating realistic eVTOL DOC.

1. Introduction

The growing population and the increasing density of economic connections lead to a growing demand for transportation. Due to the resulting emissions, however, this presents the transportation sector with challenges in terms of achieving sustainability goals (Lee et al., 2021; Barke et al., 2023; Hellmann et al., 2025). To meet the demand for transportation and to support the achievement of sustainability goals, Advanced Air Mobility (AAM) can serve as a sustainable alternative or complement to the existing ground transportation system (U.S. Government Publishing Office, 2022).

In this context, Vertical Take-off and Landing Aircraft (VTOLs) are particularly relevant, as they can take-off and land vertically, which

opens up a wide range of possible applications. Experts consider regional airport shuttle links to be among the most promising fields of application (European Union Aviation Safety Agency, 2022), but other applications such as harvesting, air ambulance services, aerial security services, and commercial air taxis are also conceivable (Goyal et al., 2018).

VTOLs differ from conventional rotorcraft due to the utilisation of more than two distributed propulsion units (European Union Aviation Safety Agency, 2024). These units can be powered by conventional fuels, electricity, hydrogen, or be operated as hybrid systems (Hader et al., 2022; Melo et al., 2022, 2023). When being powered by electricity, VTOLs are commonly referred to as Electric Vertical Take-off and Landing Aircraft (eVTOLs). Due to their unique operational capabilities

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Glossary

AAM	Advanced Air Mobility
AEA	Association of European Airlines
ATA	Air Transport Association
ATC	Air Traffic Control
CeRAS	Central Reference Aircraft Data System
DOC	Direct Operating Cost
EASA	European Union Aviation Safety Agency
eVTOL	Electric Vertical Take-off and Landing Aircraft
IOC	Indirect Operating Cost
MTOW	Maximum Take-Off Weight
MUC	Munich International Airport
NASA	National Aeronautics and Space Administration
SZG	Salzburg Airport
TOC	Total Operating Cost
VTOL	Vertical Take-off and Landing Aircraft

and their potential in terms of zero-emission operations, eVTOLs are considered an important component of AAM in the long term (Hader et al., 2022; Melo et al., 2020).

Despite the environmental benefits and operational capabilities, the widespread use of eVTOLs is largely dependent on their economic performance. Their use as airport shuttles, for example, must be economically viable and profitable for the operator. In this regard, Direct Operating Cost (DOC) is an important indicator to assess the economic performance. In the past, various methods have been developed to estimate the DOC of conventional aircraft (e.g., Maddalon, 1978; Liebeck et al., 1995; Thorbeck, 2013). However, these methods focus on conventional aircraft designs and are not applicable to eVTOLs, as the unique characteristics of eVTOLs and their operations are not accurately represented in these methods. Research dedicated specifically to the DOC estimation of eVTOLs is limited and uses only abstract, rather general approaches based on methodologies for conventional aircraft. To the best of the authors' knowledge, no publicly available method for estimating the DOC of eVTOLs exists in scientific literature.

This article aims to close this gap by introducing a novel DOC estimation model for eVTOLs, called DOCEv, providing a tool in which the DOCEv is implemented, and demonstrating its application. The DOCEv is based on the reference method of Thorbeck (2013), enables to perform a complete eVTOL performance analysis, and estimates all DOC components, specifically focusing on an accurate representation of possible eVTOL operations. Building upon an existing DOC estimation model for conventional aircraft, this article contributes to the literature by providing documentation on the first publicly available, highly specialised DOC estimation model for eVTOLs. Moreover, the model is implemented in a tool and comes with an extensive data set, enabling users to apply the DOCEv immediately.

The remainder of the article is structured as follows: Section 2 provides an overview of operation cost estimation methods in aviation. Section 3 introduces the fundamentals of airline operating cost, accompanied by a description of the developed DOCEv. The data required to apply the DOCEv is specified in Section 4. Sections 5 and 6 show the applicability of the DOCEv by analysing the use of eVTOLs as regional airport shuttles between Munich, Germany, and Salzburg, Austria, as a case study, with the results presented subsequently. The developed model is discussed in Section 7, and the article concludes in Section 8.

2. Literature Overview

For estimating aircraft operating cost, four common approaches can be applied: the analogy method, the parametric method, the engineering method, and the actual cost method (Mihara et al., 2021;

Defense Acquisition University, 2024). The analogy method (top-down method), derives the costs based on known costs of similar systems, products, or services. The parametric method uses mathematical equations representing cost estimating relationships. These describe relationships between specific parameters, such as aircraft speed, range, or weight, and historically known costs of a system, product, or service. Cost estimating relationships are based on existing aircraft data and usually defined as the best fitting curves for an investigated relationship. In the engineering method, the costs of a system, product, or service are estimated by summing up detailed cost estimations for each considered cost element (bottom-up method). Lastly, the actual cost method projects the final costs of a system, product, or service by extrapolating the costs of prototypes or performed test runs based on identified or expected trends. Most established methods estimate aircraft operating cost by a parametric cost estimation approach with multiple cost estimating relationships for different cost components. However, the approaches described can also be combined, such as in Mihara et al. (2021).

In the past, several methods focusing on the estimation of aircraft DOC were established by airlines, manufacturers, and institutions. These methods were developed for the application in comparative aircraft evaluation and are limited in scope to a single aircraft operating one (average) flight connection (Pohya et al., 2018). In general, the methods vary in the assumptions made and approaches followed, depending on the emphasis of investigation and the information available.

The first universally accepted aircraft DOC estimation method was developed by the Air Transport Association (ATA) in 1944. The ATA-method was latest revised in 1967 and still finds application today. A newer DOC estimation method taking the European airline economics into account was developed by the Association of European Airlines (AEA) in 1989. The method is split into two versions, one for short- and medium-, one for long-haul aircraft. Compared to the ATA-method, the AEA method is more detailed when it comes to engine maintenance cost and requires more input parameters (Pohya et al., 2018). Multiple DOC estimation methods were also developed by or in collaboration with the National Aeronautics and Space Administration (NASA). For example, Maddalon (1978) established a model based on the ATA method and extended it by including landing and control fees, as well as the cost of flight attendants and aircraft service. Furthermore, his method has a special focus on maintenance (Maddalon, 1978). Another NASA DOC method from Liebeck et al. (1995), also referred to as DOC+I, is based on the ATA estimation model, but extended by interest cost. In addition to that, the model includes landing and navigation fees, as well as cabin crew cost (Liebeck et al., 1995). The DOC estimation method developed by Swan and Adler (2006) is an example for a parametric method. The authors estimate the aircraft trip cost, including cost for cockpit and cabin crew, fuel, airframe maintenance, engine maintenance, and ownership, as a function, using the seat count and the trip distance as parameters. One of the most commonly applied DOC estimation methods was published by J. Thorbeck and was latest revised in 2010 (Thorbeck, 2013). The method reflects, similar to the AEA method, the economics of European airlines and was adopted as standard for the estimation of aircraft DOC in the Central Reference Aircraft Data System (CeRAS) in 2016 (Risse et al., 2016).

In the past, research has focused on the modification of existing, and partially the development of new, DOC estimation methods with respect to novel propulsion systems of conventional aircraft. Examples for this are the methods for hybrid aircraft developed by Hoelzen et al. (2018) and Vercella et al. (2020), as well as the methods for fully electric aircraft provided by Schäfer et al. (2018), Vercella et al. (2020), and Avogadro and Redondi (2024). However, these methods are developed for conventional aircraft and cannot be applied to eVTOLs, as the unique characteristics of eVTOLs and their operations are not accurately represented. Due to differences in technical design, such as Maximum Take-Off Weight (MTOW), size, or number of engines,

Table 1

Airline DOC and IOC breakdown, based on [Mensen \(2013\)](#) and [Federal Aviation Administration \(2021\)](#).

TOC		
Variable DOC	Fixed DOC	IOC
- Fuel/Energy	- Crew salary	- Administration
- Maintenance	- Aircraft ownership	- Marketing
- Fees	- Aircraft insurance	- Sales
		- Passenger service
		- Ground equipment ownership

as well as deviating operational conditions, for example regarding a potential limitation in luggages per passenger, the expected fees, or a potential battery replacement as an external service, it is necessary to add new or modify typically included cost components while discarding others. DOC estimation methods specifically for eVTOLs are scarce and use only abstract, rather general approaches based on methods for conventional aircraft. Consequently, a new method is required to accurately estimate the DOC of eVTOLs.

3. Direct Operating Cost Model for eVTOLs

3.1. Operating Cost in Commercial Aviation

The Total Operating Cost (TOC) of an airline consists of DOC and Indirect Operating Cost (IOC), as presented in Eq. (1) ([Mensen, 2013](#)).

$$TOC = DOC + IOC \quad (1)$$

These costs can be measured in various units, such as \$/year, \$/FC, \$/FH, or \$/PAX km. DOC encompasses aircraft-related costs, while IOC is more related to administration, overhead, and passengers. DOC is further distinguished into variable and fixed cost. In contrast to fixed DOC, variable DOC depends on the flight mission and the utilisation of an aircraft. Fixed DOC remains constant for different operations. [Table 1](#) breaks down airline DOC and IOC into their typical components ([Mensen, 2013](#); [Federal Aviation Administration, 2021](#)). Fuel (electric energy in the case of eVTOLs) powers the propulsion system and is distinguished into trip fuel and reserve fuel. Maintenance activities ensure the airworthiness of an aircraft and can be the inspection, repair, or replacement of degraded/life-limited parts. Maintenance costs are usually divided into labour and material costs for the engine and the airframe. The maintenance activities can be scheduled and unscheduled. Maintenance utilities, such as engineering labour, may be considered as overhead cost, but are sometimes considered as fixed DOC as well. Furthermore, operators have to pay fees for navigation, landing, and ground handling. Since fuel/energy costs, certain maintenance costs, and fees vary with the utilisation of an aircraft, they are considered as variable DOC.

Since crew salaries are expected to be paid monthly for permanent employees, they are considered part of the fixed DOC. Note that the total crew number depends on the planned utilisation. Aircraft ownership cost refers to the cost of possessing an aircraft or accessing an aircraft without necessarily purchasing it, such as through leasing. If an aircraft is purchased, depreciation and interest have to be considered. In order to finance an aircraft, operators might take a loan from a bank and pay interest. Even without financing, interest cost should be considered to reflect the cost of capital employed. If an aircraft is leased, the lease rate represents its ownership cost. In the subsequent analysis, it is assumed that the aircraft will be purchased. Not included in the aircraft ownership cost is the insurance cost against incidents and accidents. The ownership and insurance cost of an aircraft together comprise the measure of capital cost. Since the salary and the capital cost are constant with changing utilisation, they are fixed DOC. Note that the magnitude of each DOC component is influenced by several

factors, such as the aircraft, the type and scalability of the operation, the route network served, or the country in which the operation occurs.

IOC represents cost related to administration, overhead, as well as passengers, and arises from the organisation of an operator and the provision of transport service. An operator has costs for administration, marketing, and sales. Moreover, costs for passenger services and ground equipment owned are considered within the IOC ([Federal Aviation Administration, 2021](#)). In general, IOC is predominantly driven by an operator's business model ([Mihara et al., 2021](#); [Scholz, 2023](#)). As these business models are diverse and often highly specific, the proportion of IOC within the TOC may vary considerably between operators. An average TOC distribution for a commercial operator features approximately 70% DOC and 30% IOC ([EUROCONTROL, 2025](#)). DOC is consistently incurred when operating a particular aircraft and tends to be of a more similar magnitude between different operators. Therefore, the DOCEV will focus on the estimation of eVTOL DOC only.

3.2. DOCEV

3.2.1. Structure and Application

The DOCEV enables an extensive eVTOL evaluation, including an analysis of its flight performance and economics. The model is fundamentally based on the reference method of [Thorbeck \(2013\)](#), as it was adopted as standard for the estimation of aircraft DOC in the CeRAS ([Risse et al., 2016](#)). The structure of the DOCEV is modular, allowing for even small adaptations to specific areas. [Fig. 1](#) breaks down the scope of the DOCEV. In general, the model can be applied to a wide range of operational scenarios, making it valuable to multiple user groups, such as manufacturers, service providers, private or commercial operators, as well as regulatory or governmental bodies. Accurate estimates of eVTOL performance and economics can help users to define and evaluate viable business models, determine competitive pricing strategies, support fleet sizing and route planning decisions, and assess the overall economic feasibility. From the perspective of manufacturers and operators, accurate DOC estimates can, among others, support aircraft design, maintenance planning, and operations scheduling to meet market needs. For regulators and policymakers, the DOCEV can contribute to decision-making, for example, by identifying cost drivers and assessing the need for policy incentives or subsidies. Moreover, the model can support discussions on how safety, certification, or infrastructure requirements influence economic viability, market adoption, and (investment) attractiveness of eVTOL operations. Overall, the DOCEV is a comprehensive economic model that allows users to assess the economic viability and competitiveness of eVTOL operations as part of the evolving AAM ecosystem.

3.2.2. eVTOL Performance

The first module of the DOCEV offers a detailed eVTOL performance analysis for a specified flight and reserve mission based on the equations provided by [Sripad and Viswanathan \(2021a,b\)](#) and [Nathen \(2021\)](#). The performance analysis is conducted for a specified eVTOL and includes, among others, the estimation of speed, power requirements, and energy consumption for each segment of the defined flight profile (see Section 4.2). Detailed information on the mathematical formulation of the performance and energy assessment can be found in the Supporting Information S1-2.5.

3.2.3. Utilisation

The second module of the DOCEV enables an estimation of the maximum annual eVTOL utilisation as the number of annual flight cycles FC .

$$FC \left[\frac{1}{\text{year}} \right] = \text{round down} \left(\frac{OT_{p.a.} \left[\frac{\text{h}}{\text{year}} \right]}{FT [\text{h}] + BT_s [\text{h}] + TT [\text{h}]} \right) \quad (2)$$

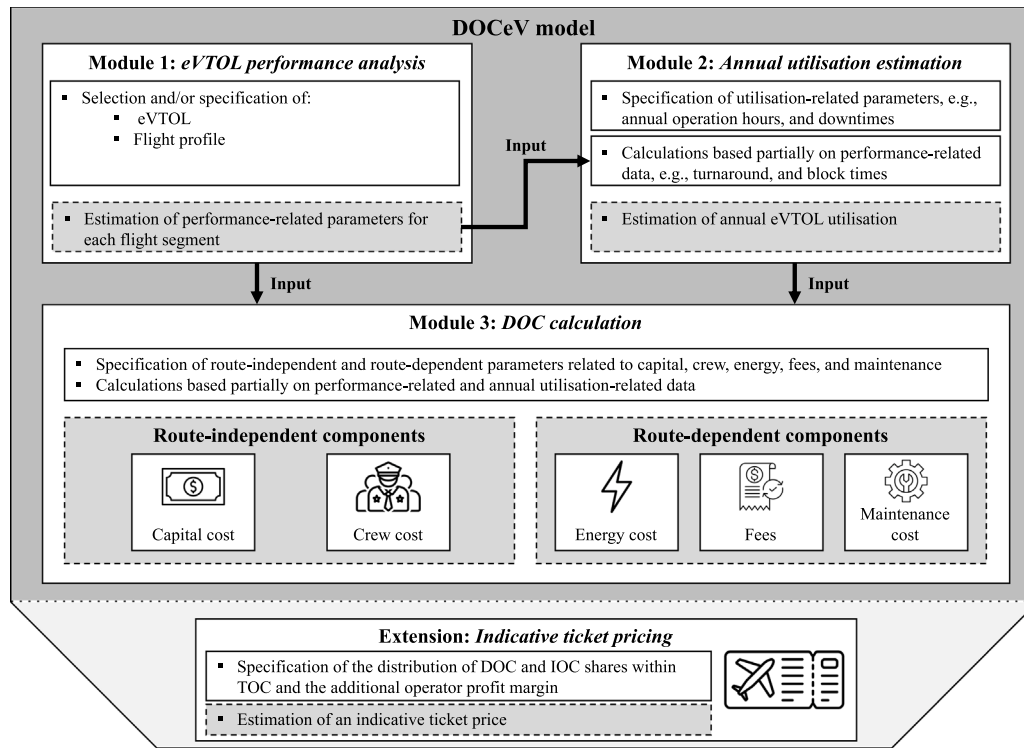


Fig. 1. Model scope of the DOCeV.

$OT_{p.a.}$ is the annual operation time of the eVTOL, which is affected by downtimes due to night curfews, maintenance events, the exceeding of certified weather conditions, crew constraints, and external factors. FT is the estimated flight time per trip and BT_s its block time supplement. The turnaround time TT is estimated by the DOCeV based on the time needed for handling, charging, boarding, de-boarding, and de-icing. To consider only complete flight cycles, calculated decimal values are rounded down to the nearest integer. The Supporting Information S1-2.2 describe the complete set of equations for the utilisation estimation.

3.2.4. Direct Operating Cost

The information from the performance analysis and annual utilisation are combined with other input data in the third module of the DOCeV. In this module, a comprehensive DOC estimation is conducted, which can be further distinguished into five sub-modules, namely the DOC components. The general structure of the DOCeV is based on the DOC estimation method for conventional aircraft developed by Thorbeck (2013). Following an extensive comparison of DOC estimation models, this method was selected as a basis due to its temporal relevance, public availability, extensive scope, few input parameters, and implementation in both the scientific literature and industry. In 2016, the method was adopted as standard for the estimation of aircraft DOC in the Central Reference Aircraft data System (Risse et al., 2016). In the DOCeV, the DOC estimation can be grouped into two parts analogously to Thorbeck's method: the route-independent, fixed cost C_1 and the route-dependent, variable cost C_2 .

$$DOC \left[\frac{\$}{\text{year}} \right] = C_1 \left[\frac{\$}{\text{year}} \right] + C_2 \left[\frac{\$}{\text{year}} \right] \quad (3)$$

The scopes of these two parts are broken down in the following, and the main differences from the reference method of Thorbeck are highlighted.

Route-independent, fixed DOC C_1

This cost includes capital cost C_{Cap} , i.e., cost related to depreciation, interest, and insurance, as well as crew cost C_{Crew} .

$$C_1 \left[\frac{\$}{\text{year}} \right] = C_{Cap} \left[\frac{\$}{\text{year}} \right] + C_{Crew} \left[\frac{\$}{\text{year}} \right] \quad (4)$$

Capital Cost C_{Cap}

As part of the capital cost C_{Cap} , the costs related to aircraft ownership are estimated, i.e., depreciation and interest, as well as the insurance cost for eVTOLs.

$$C_{Cap} \left[\frac{\$}{\text{year}} \right] = C_{AC} [\$] \cdot \left(a_{AC} \left[\frac{1}{\text{year}} \right] + f_{Ins} \left[\frac{1}{\text{year}} \right] \right) \quad (5)$$

From an investment perspective, only the aircraft purchase cost C_{AC} is considered, including the airframe, engines, and avionics, but excluding the energy system. The latter, primarily the battery, has an expected lifespan shorter than that of the eVTOL and is treated as a consumable maintenance part. f_{Ins} represents the annual insurance rate and a_{AC} an annuity factor, which accounts for the annual cost related to depreciation and interest. Unlike Thorbeck's method, the DOCeV does not differentiate between the purchase price for engines and the purchase price for the aircraft itself. More details on capital cost estimation and its mathematical formulation can be found in the Supporting Information S1-2.3.

Crew Cost C_{Crew}

As the second part of the route-independent eVTOL DOC, the DOCeV accounts for crew cost C_{Crew} .

$$C_{Crew} \left[\frac{\$}{\text{year}} \right] = b_{Crew} [-] \cdot \text{Max} (1, CI_{eq} [-]) \cdot S_{Pilot} \left[\frac{\$}{\text{year}} \right] \quad (6)$$

Unlike in the reference method of Thorbeck, only the salary of a pilot S_{Pilot} is considered. Salaries of flight attendants are not included. Yet, the DOCeV incorporates a crew index equivalent CI_{eq} , representing the minimum number of pilots required per eVTOL. This equivalent is calculated considering the annual flight cycles, the block and turnaround

time of one flight mission, as well as an annual pilot productivity. For future autonomous eVTOL operations, the DOCeV also allows to exclude cockpit crew salaries with the binary variable b_{Crew} . The detailed mathematical formulation for the cost estimation of the crew is presented in the Supporting Information S1-2.4.

Route-dependent, variable DOC C_2

This cost encompasses the costs for energy and fees for one trip, C_{Energy} and C_{Fees} , as well as the maintenance costs per flight cycle C_{Mtn} . The annual flight cycles for the flight connection considered are denoted as FC .

$$C_2 \left[\frac{\$}{\text{year}} \right] = FC \left[\frac{1}{\text{year}} \right] \cdot (C_{Energy} [\$] + C_{Fees} [\$] + C_{Mtn} [\$]) \quad (7)$$

Energy Cost C_{Energy}

To estimate the energy cost per trip C_{Energy} , the final step in the calculation varies only little from Thorbeck's reference method, since not the consumed fuel is multiplied by the energy price P_{Energy} , but the consumed trip and reserve energy, E_{Trip} and $E_{Reserve}$, respectively.

$$C_{Energy} [\$] = P_{Energy} \left[\frac{\$}{\text{kWh}} \right] \cdot (E_{Trip} [\text{kWh}] + E_{Reserve} [\text{kWh}]) \quad (8)$$

The estimation of the total energy consumption, including the trip and reserve energy, is part of the eVTOL performance analysis and carried out for a user-specified eVTOL operating a user-defined flight mission. Here, the DOCeV also includes binary variables, allowing to include or exclude three energy reserves introduced by the authorities: the contingency reserve, the alternate reserve, and the final reserve (European Union Aviation Safety Agency, 2022). In general, the energy consumption is calculated for each flight segment separately by multiplying an estimated required power by the approximated flight time in this segment. In order to estimate the flight time, the speed and distances flown during each flight segment are estimated, if not specified by the user. More information on the individual eVTOL performance analysis of each flight segment and the energy cost estimation are provided in the Supporting Information S1-2.5.

Fees C_{Fees}

Another route-dependent DOC component is fees per flight cycle C_{Fees} . The fees are further distinguished into handling, navigation, and landing fees, $C_{Handling}$, $C_{Navigation}$, and $C_{Landing}$, respectively.

$$C_{Fees} [\$] = C_{Handling} [\$] + C_{Navigation} [\$] + C_{Landing} [\$] \quad (9)$$

In contrast to the reference method of Thorbeck, which uses a very general approach to estimate the handling fees $C_{Handling}$, the DOCeV specifically focuses on the cost estimation of handling processes, relevant to eVTOL operations.

$$C_{Handling} [\$] = C_{Luggage} [\$] + C_{Ground} [\$] + C_{Clean} [\$] + C_{De-Ice} [\$] \quad (10)$$

First, additional costs are considered for the transportation of bulky luggage $C_{Luggage}$, since eVTOLs are expected to have only limited space available for luggage. Assuming a pilot and passenger weight of 75 kg per person, the total allowed luggage capacity per passenger is likely to fall within the range of 15 kg to 19.5 kg (Lilium, 2024b; Joby Aviation, 2023). Considering a potential eVTOL launch service as (regional) airport shuttle, the storage space available on board the eVTOLs may not be sufficient to carry all luggage. Hence, it is assumed that eVTOLs provide only enough space for each passenger's carry-on luggage, with heavier pieces of luggage being transported separately. Whether these associated fees belong to the DOC or passenger-related IOC is debatable, but the DOCeV can account for this cost. In addition to the cost of luggage transportation, the DOCeV also considers a fee for the use of the ground infrastructure C_{Ground} , such as eVTOL charging stations or battery cooling systems. In addition, handling fees for periodic interior cleaning C_{Clean} and eVTOL de-icing C_{De-Ice}

are taken into account. As for Thorbeck's method, the navigation fees $C_{Navigation}$ are estimated following the published calculation formula of the Central Route Charge Office of EUROCONTROL (2022).

$$C_{Navigation} [\$] = UR_{Nav} [\$] \cdot \frac{R_{Nav,eff} [\text{km}]}{100 \text{ km}} \cdot \sqrt{\frac{MTOW [\text{kg}]}{50,000 \text{ kg}}} \quad (11)$$

This approach is extended by the integration of a factor $R_{Nav,eff}$ that represents the actual share of flight distance through controlled airspace while having Air Traffic Control (ATC) contact. This is necessary, as, for example, in Germany, flights under visual flight rules can be carried out with or without ATC contact up to the end of airspace class E, i.e., up to flight level 100 over flat ground and flight level 130 over the Alps (German Federal Bureau of Aircraft Accident Investigation, 2011). UR_{Nav} is the applicable service unit rate defined for each country in Europe and $MTOW$ represents the maximum take-off weight.

The last type of fees considered in the DOCeV, the landing fees $C_{Landing}$, includes charges for processes and services in and around the take-off and landing sites, the so-called vertiports.

$$C_{Landing} [\$] = \frac{P_{Landing,A} [\$] + P_{Landing,B} [\$]}{2} \quad (12)$$

The landing fees $P_{Landing,A/B}$ for departure (A) and arrival destination (B) are expected to be determined in the vertiports' charge tables. As landing fees can vary from vertiport to vertiport, it can occur that an eVTOL takes off at a less expensive vertiport and lands at a more expensive vertiport. Therefore, the landing fees considered in the DOCeV are the arithmetic mean of the landing fees at both vertiports. In Thorbeck's reference method, landing fees are estimated based on the aircraft MTOW only. More detailed information on the three types of fees considered in the DOCeV and their mathematical formulation can be found in the Supporting Information S1-2.6.

Maintenance Cost C_{Mtn}

As the last route-dependent DOC component, eVTOL maintenance cost C_{Mtn} , is estimated in the DOCeV. The applied approach is different from the parametric cost estimation approach presented in the reference method of Thorbeck. The DOCeV is able to consider either cost of in-house maintenance $C_{InHouse,Mtn}$ or subcontracted maintenance $C_{SubCon,Mtn}$.

$$C_{Mtn} [\$] = \frac{1}{FC \left[\frac{1}{\text{year}} \right]} \cdot \left(b_{InHouse,Mtn} [-] \cdot C_{InHouse,Mtn} \left[\frac{\$}{\text{year}} \right] + b_{SubCon,Mtn} [-] \cdot C_{SubCon,Mtn} \left[\frac{\$}{\text{year}} \right] \right) \quad (13)$$

FC represents the annual flight cycles. $b_{InHouse,Mtn}$ and $b_{SubCon,Mtn}$ are binary variables. The estimation approach for the in-house maintenance cost is based on Sieb et al. (2023) and was substantially extended. In general, the in-house maintenance cost encompasses the labour and material costs related to eVTOL line and base maintenance $C_{Mtn,eVTOL}$, unscheduled maintenance cost $C_{Mtn,unsched}$, as well as overhead cost $C_{Mtn,Overhead}$.

$$C_{InHouse,Mtn} \left[\frac{\$}{\text{year}} \right] = C_{Mtn,eVTOL} \left[\frac{\$}{\text{year}} \right] + C_{Mtn,unsched} \left[\frac{\$}{\text{year}} \right] + C_{Mtn,Overhead} \left[\frac{\$}{\text{year}} \right] \quad (14)$$

Line maintenance is expected to be performed before or after every flight cycle. The base maintenance is split into two parts, one for the energy system, and one for the remaining eVTOL. Energy system base maintenance events occur once an energy system reaches its end-of-life. The DOCeV allows to select whether a new energy system is purchased or provided externally as a service. The base maintenance events considering the remaining eVTOL follow an alternating flight time driven, as well as flight cycle driven maintenance plan, which

is specified by the user. Moreover, the user can define the scope and, therefore, the cost of each maintenance event. In addition to this scheduled maintenance, the DOCeV takes unscheduled maintenance into account. The cost of unscheduled maintenance $C_{Mtn,unsched}$ is estimated based on the scheduled maintenance cost and an ageing factor. Finally, the DOCeV includes the cost associated with maintenance overhead $C_{Mtn,Overhead}$. In particular, the costs related to slack labour, hangar rent, component supply, IT systems, and a continuing airworthiness management organisation are included in the model.

The annual costs of a subcontracted maintenance are assumed to be based on a fixed component $C_{SubCon,Mtn,fix}$ and a variable component $C_{SubCon,Mtn,var}$ that depends on the eVTOL utilisation. The utilisation is defined by the number of annual flight cycles FC and the flight time per cycle FT .

$$C_{SubCon,Mtn} \left[\frac{\$}{year} \right] = C_{SubCon,Mtn,fix} \left[\frac{\$}{year} \right] + C_{SubCon,Mtn,var} \left[\frac{\$}{h} \right] \cdot FC \left[\frac{1}{year} \right] \cdot FT [h] \quad (15)$$

The detailed description of the maintenance cost estimation in the DOCeV is provided in Supporting Information S1-2.7.

The general mathematical formulation of the DOCeV consists of 298 variables and parameters across 171 equations. Due to the modular structure of the DOCeV, the model is easy to modify and, therefore, suitable for application to a wide range of potential eVTOL operations. This article is accompanied by an implementation of the DOCeV in Microsoft Excel version 16.77.1. In the model, each input can be specified individually by the user. Furthermore, users can define own eVTOL designs and flight missions according to their context of investigation.

3.2.5. Indicative Ticket Pricing

Based on the estimated DOC, the DOCeV provides an extension that enables the indicative estimation of the eVTOL ticket prices. The estimation approach used follows the approach of Niklaß et al. (2020) and first estimates the TOC based on a user-defined share between DOC and IOC. Afterwards, the total required revenue is calculated by adding an eVTOL operator profit margin. Fig. 2 illustrates the approach. The final indicative ticket prices can be obtained in various units, for example, by dividing the revenue by the average seat occupation, i.e., the number of seats multiplied by an average passenger load factor. Note that this approach represents only a rough estimation rather than a precise ticket price calculation. The described method is not intended to compete with the classical revenue management, as it cannot capture the full complexity of such systems. Moreover, specific ticket pricing structures, including, for example, a base fare and a distance-dependent fare, are not accurately depicted by this approach. Instead, it merely provides potential indicative ticket prices to support the user's interpretation of results.

4. DOCeV Data Requirements

For applying the DOCeV, an extensive data set is required. Such an exemplary data set is provided in the related Supporting Information, as well as implemented in the Excel tool. This input data is based on values derived for an eVTOL operation as a regional airport shuttle in Germany. Consequently, the input values of certain parameters may need to be adapted when applying the DOCeV to a different operational scenario or an operation located in a different geographical area. The general types of input data required for the model application are described in the following subsections. They are divided into eVTOL characteristics, flight mission, eVTOL utilisation, as well as route-independent and route-dependent components. The entire input data set is presented in the Supporting Information S1-3.

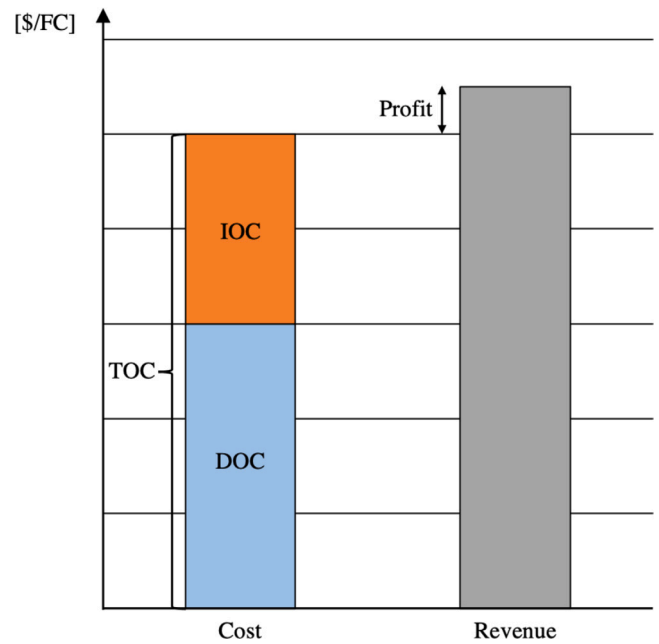


Fig. 2. Schematic approach for estimating the revenue per trip, according to Niklaß et al. (2020).

4.1. eVTOL Characteristics

The eVTOL input parameters can be categorised into aerodynamic, configurational, and operational parameters. Aerodynamic input data comprises the eVTOL design characteristics, such as engine design, disk area, and wing surface area. In addition, aerodynamic and efficiency metrics, including the zero-lift drag coefficient, the maximum lift-to-drag ratio, as well as vertical, horizontal, and vectorial flight efficiencies, are required.

Configurational input data includes the MTOW, the maximum payload mass, and the number of passenger seats. Furthermore, the energy system must be specified, detailing the number of system units, their individual weight, and the total energy storage capacity.

Lastly, operational input data for the eVTOL is required. Specifically, input values must be provided for the average passenger load factor, the cruise speed, the on-board power consumption, and the rolling resistance coefficient during taxiing.

4.2. Flight Mission

The vertical flight profile considered is orientated at Uber (2018) for the regular trip and at the requirements specified by European Union Aviation Safety Agency (2022) for the reserve mission. A schematic vertical flight profile, including the regular trip and the reserve mission, is shown in Fig. 3. The regular mission consists of eleven flight segments and is illustrated in black. Furthermore, reserve energy must be considered. Here, the European Union Aviation Safety Agency (EASA) defined three types of reserves for eVTOLs: contingency, alternate, and final reserve. The contingency reserve is required to compensate for unforeseen factors during the trip, such as headwind, weather detours, or delays in the air or on the ground. The alternate reserve is required to have the option for a flight to an alternate destination, e.g., when circumstances prevent a landing at the intended destination. Note that since 2023, additional energy for an alternate mission has no longer been explicitly required (European Union Aviation Safety Agency, 2023). In addition to the contingency and alternate reserve, the EASA defines a final reserve. This final reserve energy has to

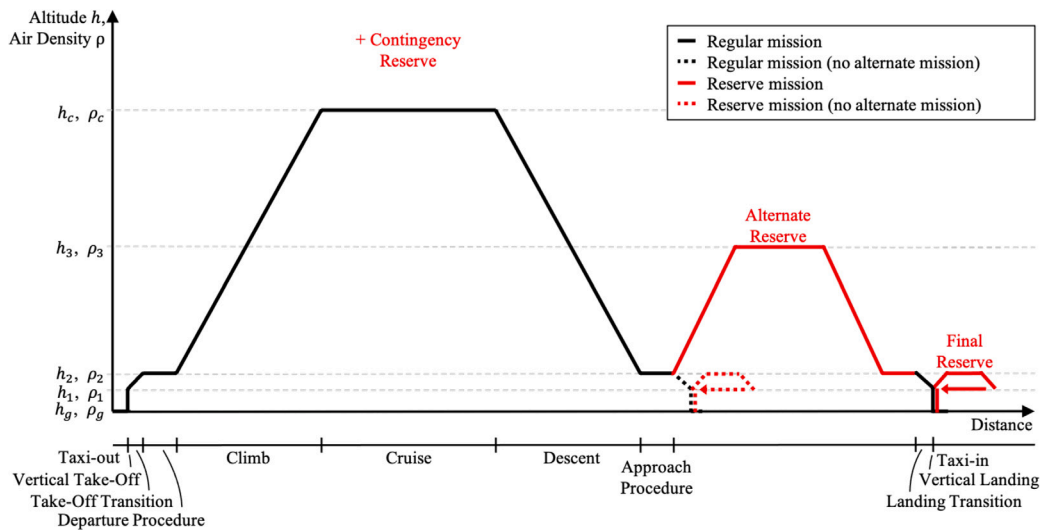


Fig. 3. Considered vertical flight profile, according to Uber (2018) and European Union Aviation Safety Agency (2022).

be sufficient for the eVTOL to perform a go-around and another approach (European Union Aviation Safety Agency, 2022). In the DOCeV, this is interpreted as the energy needed to perform another take-off transition, followed by a short horizontal go-around flight, a re-transition, and a vertical landing. The reserves are coloured in red in Fig. 3.

In the DOCeV, all altitudes and corresponding air densities indicated in Fig. 3 are required as input. Furthermore, the vertical speed of each relevant flight segment and the speed during taxiing must be defined. Lastly, the DOCeV requires input data for the taxiing distance, the horizontal distances of the departure and approach procedure, the total great circle distance between the departure and the arrival vertiport, the total alternate distance, as well as the horizontal go-around distance during a final reserve mission. Since eVTOLs will most likely not be able to fly the great circle distance between departure and arrival destinations, the horizontal flight path is considered in the DOCeV by including a detour factor. This detour factor is composed of a fixed distance, accounting for standardised departure and arrival procedures, as well as a variable detour factor.

4.3. eVTOL Utilisation

The maximum annual utilisation of an eVTOL is estimated in the DOCeV as the maximum number of annual flight cycles. This calculation requires input for several parameters, including the potential annual operating hours, the number of operational days per year, the annual eVTOL downtime, and the turnaround time per flight cycle. Regarding downtime, the annual number of hours on the ground due to factors such as night curfews at airports, maintenance activities, weather conditions exceeding certified operational thresholds, and crew availability constraints are required as input. The turnaround time per flight cycle is approximated based on the specified durations for ground handling, boarding and de-boarding procedures, as well as eVTOL charging and de-icing. Additionally, the annual share of flights that require an eVTOL de-icing prior departure must be specified. Furthermore, the block time is incorporated into the estimation of the eVTOL utilisation. This parameter is derived as part of the eVTOL performance analysis described in Section 3.2.

4.4. Route-independent Components

To estimate the route-independent DOC associated with capital cost and crew, various input parameters regarding these two DOC

components are required in the DOCeV. For capital cost estimation, the purchase price of an eVTOL in relation to its operational empty weight (including components such as engines and avionics, but excluding the energy system) must be specified. In addition, the annual insurance rate, the annual interest rate, the residual value factor, and the depreciation period of an eVTOL are required.

For the crew cost estimation, the DOCeV requires a determination of whether the operations will be autonomous or piloted. If piloted operations are considered, it is necessary to define the annual salary of a cockpit crew member and the annual net pilot productivity. The latter represents the actual productive hours of a pilot per year, taking into account factors such as training, vacation, and other non-productive periods.

4.5. Route-dependent Components

The DOCeV requires various input data to estimate the energy cost, fees, and maintenance cost associated with an eVTOL operation. Estimating the energy cost requires a decision regarding the inclusion or exclusion of the contingency, alternate, and/or final reserves. If the contingency reserve is to be included, its magnitude as a share of the total trip energy must be provided as input, along with the annual proportion of flights that use this reserve energy. Similarly, the definition of the annual proportion of flights that utilise an alternate and final reserve is required. Furthermore, the energy price per kilowatt-hour must be provided.

Estimation of total fees per flight cycle requires various input pertaining to handling, navigation, and landing fees. Within the scope of handling fees, a ground infrastructure utilisation fee must be specified, representing the use of ground systems, such as charging infrastructure or battery cooling systems. This fee excludes passenger-related infrastructure, as these costs are included in landing fees. Furthermore, the cost of transporting luggage (other than carry-on) using other means of transportation is required as input, as well as the average number of such items per passenger. In addition to these, the cost associated with an eVTOL interior cleaning, together with the average number of cleaning sessions per day, is required. As the final component of ground handling fees, the cost of de-icing an eVTOL must be specified. To estimate navigation fees, the DOCeV requires input in terms of the en-route navigation service unit rate and the anticipated proportion of flight distance carried out within controlled airspace while having ATC contact. Additional input, such as flight distance, is derived from the definition of the flight mission. Estimation of landing fees necessitates

only the specification of landing fees at both the departure and arrival vertiport.

To estimate the maintenance cost, the DOCeV requires a decision of whether subcontracted or in-house maintenance should be considered. For subcontracted maintenance, annually fixed maintenance cost as well as flight hour dependent, variable maintenance cost must be defined. If in-house maintenance is considered, different input parameters are required. First, the mechanic's labour rate as well as the man hours and material cost of a single line maintenance event must be specified. In addition, the energy system base maintenance needs to be defined. Here, the purchase of an energy system or its provision as a service can be considered, once the energy system, i.e., the battery, reaches its end of life. Depending on the investigated scenario, input is required either in terms of the energy system purchase price per kilowatt-hour (including the battery and ancillary components, excluding engines) and the associated man hours for a replacement, or the annually fixed and variable costs for the external provision of the energy system. In all cases, a minimum battery state-of-health and average battery degradation per flight cycle must be defined. To estimate the base maintenance cost for the remaining eVTOL, mean-flight-hours-between-maintenance and mean-flight-cycles-between-maintenance are required as input. Given that the eVTOL base maintenance is assumed to alternate between minor and more extensive scopes for both the flight hour driven and the flight cycle driven maintenance events, the DOCeV necessitates the specification of the required man hours and material cost for each of these four distinct maintenance scenarios. To account for unscheduled maintenance cost as a share of the scheduled base maintenance cost, an age-independent and an age-dependent share have to be specified in addition to the considered eVTOL age. Finally, overhead costs are included. For their estimation, input is required regarding the number of mechanics per eVTOL, an effective labour rate of a mechanic during slack periods, and the annual working hours of a mechanic. Furthermore, additional overhead costs for hangar and office rent, component supply, IT systems, and annual costs for a continuing airworthiness management organisation may also be defined.

5. Case Study: Airport Shuttle between Munich International Airport (MUC) and Salzburg Airport (SZG)

To demonstrate the applicability of the DOCeV and validate its accuracy, the model is applied to a case study. To do this, a potential eVTOL operation serving as a regional airport shuttle between Munich International Airport (MUC) and Salzburg Airport (SZG) is investigated. Experts consider this regional airport shuttle to be among the most promising eVTOL application scenarios (EIT Urban Mobility, 2022). According to Schunck et al. (2024), Munich is the city with the highest viability and feasibility for AAM operations in Europe. Furthermore, Salzburg is considered a high-income region in Europe (Statistics Austria, 2022). This may become important as initial eVTOL operations will most likely target premium passengers (Biehle, 2022).

The great circle distance between the two airports is 109 km, which is 62% shorter than the driving distance by car (Georg, 2024b), as illustrated in Fig. 4. Even when applying a detour factor of 24% (see Supporting Information S1-3.2), the total flight distance of 135 km would still be approximately 30% shorter. On average, the travel time by car from MUC to SZG is between 1 h 40 min and 2 h 30 min (Google Maps, 2024). An alternative train journey between MUC and Salzburg main station takes between 2 h 15 min and 2 h 50 min, with a connecting bus to SZG requiring an additional 22 min (Deutsche Bahn, 2024; Salzburger Verkehrsverbund GmbH, 2024). In contrast, the flight time of a helicopter operation between MUC and SZG is approximately 30 min (SennAir Helicopter, 2024). However, it is reasonable to add 20 min to this flight time to account for passenger processes at the departure site, such as check-in and security control (GlobeAir, 2024). For the airport shuttle operation, the six-seater Lilium Jet and the four-seater Joby Aviation S4 are considered eligible eVTOLs, based on their

performance characteristics (Lilium, 2024a; Joby Aviation, 2021). Both eVTOLs are projected to reach a maximum operational range of up to 250 km. Moreover, the Joby Aviation S4 is supposed to operate at a maximum cruise speed of approximately 320 km/h, while the Lilium Jet can fly up to 250 km/h during cruise. In addition, comprehensive data on the design and potential usage patterns are available for both eVTOLs.

The calculation of the DOC is performed using the DOCeV, implemented in Microsoft Excel version 16.77.1. The input data is presented in Supporting Information S1-3. Note that the number of bulky luggage per passenger is assumed to be zero. Therefore, no additional luggage transportation cost is considered. This is due to the fact that it is debatable whether luggage transportation falls under DOC or passenger-related IOC. Furthermore, the contingency, alternate, and final reserve missions are excluded in the DOC analysis. This aligns with most current DOC estimation methods, which typically do not include reserves when estimating aircraft DOC. Moreover, it is assumed that eVTOLs will be operated by pilots. It is presumed that the maintenance is performed in-house, and that batteries are purchased as consumable components rather than being provided as a service. In addition, expenses for component supply, maintenance IT, and a continuous airworthiness management organisation are set to zero in the overhead cost to avoid the influence of incorrect input data.

6. Results

In the following, the annual DOC, the DOC per flight cycle, and the DOC per flight hour will be calculated using the units \$/year, \$/FC, and \$/FH. Note that in the mathematical formulation of the DOCeV, the latter two units are denoted as \$ and \$/h. The alternative notations using \$/FC and \$/FH, which are also applied in the calculation and results worksheets of the implemented tool, are supposed to facilitate reading and interpretation.

6.1. DOC Estimates for a Lilium Jet

The flight time of a Lilium Jet for one 135.35 km mission between MUC and SZG is estimated to be 0.57 h. The block time supplement is 0.01 h. Given the input parameters, an approximated turnaround time of 0.98 h is obtained. Based on these estimated time periods and the assumed potential annual operating hours, the maximum number of annual flight cycles is calculated as 2787.

The annual DOC for a Lilium Jet operating between MUC and SZG is estimated to be 3,004,137 \$/year. Given the estimated annual utilisation and flight time, the DOC can also be stated as 1078 \$/FC or 1887 \$/FH. The DOC breakdown is presented in Table 2. The findings indicate that maintenance-related costs constitute the largest proportion of the DOC, amounting to approximately 29%. As can be seen, maintenance costs are mainly driven by labour and material costs of the energy system base maintenance. The remaining labour and material costs for the line maintenance and the base maintenance of the eVTOL are estimated to be considerably lower. The handling, navigation, and landing fees account for 29% of the DOC as well. The spread observed between these fees is wide: navigation fees constitute just 1% of the DOC, while landing fees alone account for approximately 20%. The capital costs, including all costs associated with interest, depreciation, and insurance, account for 24% of the DOC. The estimated crew costs represent 13% of the DOC and the energy costs 6%. Overall, route-independent costs account for a smaller share of the DOC (37%) than route-dependent costs (63%).

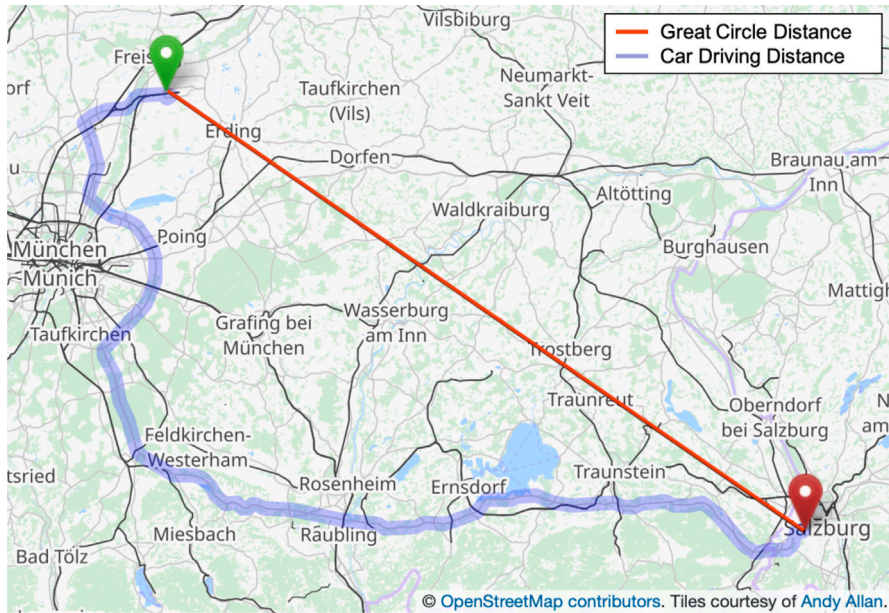


Fig. 4. Great circle distance (red) and car driving distance (blue) for a trip from MUC to SZG, based on OpenStreetMap (2025). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Detailed breakdown of the estimated DOC for the use of a Lilium Jet operating as an airport shuttle between MUC and SZG.

	\$/year	\$/FC	\$/FH
Crew (C1)	392,741	141	247
Capital (C1)	706,451	253	444
Energy (C2)	194,697	70	122
In-house Maintenance (C2)	863,256	310	542
Line Mtn. (L&M)	79,921	29	50
Base Mtn. (L&M) excl. ESys	83,592	30	53
ESys Base Mtn. (L&M)	520,344	187	327
Unscheduled Mtn.	83,431	30	52
Overhead	95,969	34	60
Fees (C2)	846,992	304	532
Handling	230,491	83	145
Navigation	29,083	10	18
Landing	587,419	211	369
DOC	3,004,137	1078	1887
C1 (route-independent)	1,099,192	394	690
C2 (route-dependent)	1,904,945	684	1197

Mtn. = Maintenance; L&M = Labour and Material; ESys = Energy System.

Table 3
Detailed breakdown of the estimated DOC for the use of a Joby Aviation S4 operating as an airport shuttle between MUC and SZG.

	\$/year	\$/FC	\$/FH
Crew (C1)	392,836	94	193
Capital (C1)	583,930	140	286
Energy (C2)	149,534	36	73
In-house Maintenance (C2)	807,882	194	396
Line Mtn. (L&M)	119,294	29	58
Base Mtn. (L&M) excl. ESys	113,727	27	56
ESys Base Mtn. (L&M)	393,911	95	193
Unscheduled Mtn.	76,486	18	38
Overhead	104,464	25	51
Fees (C2)	1,137,654	273	558
Handling	309,499	74	152
Navigation	37,773	9	19
Landing	790,382	190	387
DOC	3,071,837	738	1506
C1 (route-independent)	976,767	235	479
C2 (route-dependent)	2,095,070	504	1027

Mtn. = Maintenance; L&M = Labour and Material; ESys = Energy System.

6.2. DOC Estimates for a Joby Aviation S4

The flight time of a Joby Aviation S4, which is estimated to be 0.49 h, is shorter than that of the Lilium Jet. This results from the higher cruise speed of the Joby Aviation S4. The block time supplement of 0.01 h is the same for both eVTOLs. Since the Joby Aviation S4 consumes less energy and has fewer seats, the turnaround time is estimated to be only 0.54 h, almost half of what is estimated for the Lilium Jet. The maximum annual flight cycles of the Joby Aviation S4 are calculated to be 4160, that is 49% more compared to those of the Lilium Jet. The estimated DOC for the Joby Aviation S4 is broken down at the component level in Table 3.

The annual DOC for operating the Joby Aviation S4 as a regional airport shuttle between MUC and SZG is estimated to be 3,071,837 \$/year. The fees account for the largest proportion of the DOC, amounting to approximately 37%. This proportion is driven by landing fees, representing 26% of the DOC. In comparison, the share of all maintenance-related costs is 26%. Within the maintenance costs, the labour and material costs of the energy system base maintenance account for the

largest share of the DOC with 13%. The capital costs represent 19% of the DOC. The cockpit crew costs account for 13% of the DOC and the energy costs for 5%.

6.3. Comparison of the DOC per Flight Cycle

The estimated annual DOC of the Lilium Jet and the Joby Aviation S4 are of the same magnitude. However, when assuming the relatively high cruise speed claimed by Joby Aviation, the S4 is projected to fly more flight cycles annually and require less flight time per trip. Consequently, the estimated DOC per flight cycle of 738 \$/FC for the Joby Aviation S4 is 31% lower than that for the Lilium Jet. The DOC per flight hour of 1506 \$/FH is 20% lower compared to the Lilium Jet. Fig. 5 compares the DOC per flight cycle for the Joby Aviation S4 to that of the Lilium Jet.

As illustrated in Fig. 5, significant differences in the DOC per flight cycle are driven by capital costs, crew costs, and base maintenance costs of the energy system. Due to a higher purchase price (excluding the

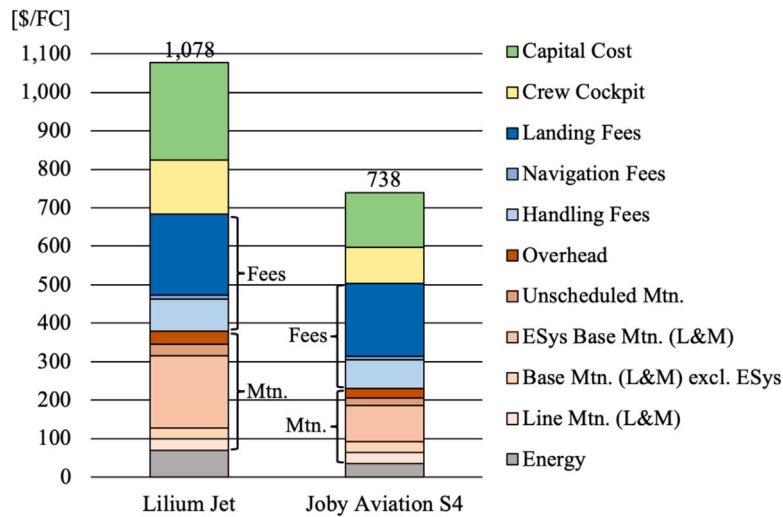


Fig. 5. Comparison of DOC per flight cycle for the Lilium Jet and Joby Aviation S4 operating between MUC and SZG.

battery) of the Lilium Jet and the fewer anticipated annual flight cycles, the capital costs per flight cycle are lower for the Joby Aviation S4. The crew costs per flight cycle are estimated to be higher for the Lilium Jet, as its flight time and turnaround duration are estimated to be 14% and 45% longer than those of the Joby Aviation S4. Since eVTOL pilots are expected to work during both the flight and the turnaround in the DOCEV, the duration of these phases are counted as working hours. Therefore, the Lilium Jet requires a greater number of pilot working hours to complete a full flight cycle, including turnaround, resulting in higher cockpit crew costs per flight cycle. Besides crew costs, the energy system base maintenance costs are estimated to be considerably higher for the Lilium Jet. The degradation of batteries is projected to be very similar for both eVTOLs in the DOCEV. Consequently, the batteries require replacement after approximately the same number of flight cycles performed. Given that the acquisition cost of Lilium's battery are estimated to be 110% higher than those of Joby Aviation's battery, the energy system base maintenance costs for the Lilium Jet are higher. The next largest difference in the DOC per flight cycle arises from the different energy costs, as the higher energy consumption of the Lilium Jet causes higher DOC per flight cycle. All other DOC components not mentioned differ only marginally between the two considered eVTOLs.

6.4. Indicative Ticket Pricing and Comparison with other Modes of Transportation

To provide an alternative perspective on the results, indicative eVTOL ticket prices for the Lilium Jet and Joby Aviation S4 are estimated based on their projected DOCs, following the approach of Niklaß et al. (2020). Given that the proportions of DOC and IOC in the total TOC, along with operator profit margins, can vary significantly between operators, the estimated ticket prices for a regional airport shuttle between MUC and SZG are presented for both Lilium Jet and Joby Aviation S4 in Tables 4 and 5 in a matrix format. The investigated Lilium Jet provides six passenger seats, while the Joby Aviation S4 can hold up to four passengers (Lilium, 2024a; Joby Aviation, 2021). Moreover, an average passenger load factor of 76.1% is assumed (International Air Transport Association, 2024). Estimated ticket prices for the Lilium Jet are generally slightly lower than those for the Joby Aviation S4. The ticket prices vary widely depending on the cost and profit margin scenario, ranging from 354\$/PAX to 558\$/PAX. Note that these ticket prices do not consider the opportunity of single on-demand trips and that the estimated underlying DOC excludes cost for additional transportation of bulky luggage.

Compared to existing ground transportation, the estimated eVTOL ticket prices are considerably more expensive. For instance, a train

Table 4

Indicative ticket prices per passenger under different cost and profit margin scenarios for the Lilium Jet operating between MUC and SZG.

[/PAX]		Operator profit margin		
		5%	10%	15%
DOC : IOC	50 : 50	496	519	543
	60 : 40	413	433	452
	70 : 30	354	371	388

Assumptions: Passenger Seats = 6, Average Passenger Load Factor = 76.1%.

Table 5

Indicative ticket prices per passenger under different cost and profit margin scenarios for the Joby Aviation S4 operating between MUC and SZG.

[/PAX]		Operator profit margin		
		5%	10%	15%
DOC : IOC	50 : 50	509	534	558
	60 : 40	425	445	465
	70 : 30	364	381	399

Assumptions: Passenger Seats = 4, Average Passenger Load Factor = 76.1%.

and bus ticket booked one month in advance costs around 37\$/PAX (Deutsche Bahn, 2024; Salzburger Verkehrsverbund GmbH, 2024). Alternatively, a taxi ride shared by three passengers for the driving distance of 176.79 km between MUC and SZG costs approximately 173\$/PAX (Georg, 2024b; Muenchen.de, 2024). Compared to helicopter transportation, the eVTOL ticket prices may be competitive. For example, a ticket for a helicopter flight between London Gatwick Airport and Oxford Airport, which is the same great circle distance as MUC to SZG, costs 1426\$/PAX when booking one month in advance (Georg, 2024a; London Heliport, 2024).

Although the estimated ticket prices for eVTOL transportation are high compared to transportation by taxi or train, eVTOL operation between MUC and SZG comes with the advantage of significant savings in travel time. The eVTOL block time, estimated at approximately 30 min, can be extended by another 25 min to account for passenger processes at the departure and arrival vertiport. However, the total travel time of an eVTOL operation between MUC and SZG is still approximately 1 h to 2 h shorter than the journey by taxi or train. Besides the advantage of shorter travel time, eVTOL transportation has the opportunity to optimise the travel journey due to on-demand operations that minimise the connection time to other flights.

Overall, eVTOL transportation offers advantages in terms of travel time savings and travel journey optimisation. However, the ticket prices

are significantly higher compared to ground transportation. In addition, travel comfort may be reduced if passengers and/or luggage require additional transfers. Consequently, a passenger's choice of transportation mode is expected to depend significantly on their individual priorities and preferences.

7. Discussion

7.1. Classification of the Results

In this article, the DOCEv, a model for the estimation of eVTOL DOC, has been developed, implemented, and applied to a case study, using an established set of input data. Generally, the estimates provided by the DOCEv in the context of the case study are considered realistic and suitable for further investigation. However, comparing the estimated DOC for the six-seater Lilium Jet and the four-seater Joby Aviation S4 with the findings of the literature is not trivial, as the DOC depends on several factors. Due to the fact that many literature findings are presented in the unit \$/PAX km, this unit will be used for the comparison. The results of the DOCEv are normalised using the great circle distance of the considered connection between MUC and SZG. Additionally, an average passenger load factor of 76.1% is assumed (International Air Transport Association, 2024). It is observed that the estimated DOC per passenger-kilometre increases for shorter flight distances and decreases for longer flight distances. One of the primary reasons for this is the presence of DOC components that are independent of the flight distance. However, this distance-dependent behaviour may not always be obvious, as the DOCEv automatically adapts the maximum number of annual flight cycles when changing the flight distance. A reduction in flight distance leads to an increase in the number of annual flight cycles. Therefore, the DOC per flight cycle would decrease, while the DOC per passenger-kilometre would increase. Note that users can always overwrite the automatic estimation of annual flight cycles in the Excel tool.

The DOC per passenger-kilometre for the Lilium Jet operating between MUC and SZG is estimated to be 2.17 \$/PAX km. When reducing the great circle distance from 109 km to 80 km, the DOC per passenger-kilometre increases to 2.65 \$/PAX km. For the Joby Aviation S4 that operates between MUC and SZG, the DOC per passenger-kilometre is 2.23 \$/PAX km. On an 80 km great circle distance mission, the corresponding DOC per passenger-kilometre amounts to 2.80 \$/PAX km. The DOC per passenger-kilometre reported in the literature show variations. Goyal et al. (2018) estimated DOC of 2.46 \$/PAX km for a five-seater eVTOL performing an 80 km mission. Based on the projected ticket prices provided by Niklaß et al. (2020), the DOC for a four-seater eVTOL performing an 80 km flight mission can be approximated to be 2.80 \$/PAX km. Kohlman et al. (2019) estimated the DOC for a four-seater eVTOL on a 20 min flight mission to be 3.37 \$/PAX km. A potential advancement in battery technology could reduce the estimated DOC to 3.04 \$/PAX km. For a 40 km mission, Ploetner et al. (2020) estimated the DOC for a two-seater eVTOL to be 3.18 \$/PAX km. Pertz et al. (2023) explored even shorter flight distances, estimating the DOC for a four-seater Archer Aviation Midnight on a 15 km mission to be 4.16 \$/PAX km. Additionally, Pertz et al. (2023) projected the DOC for a six-seater Lilium Jet operating a 186 km mission between Hamburg and Sylt. The authors estimated the DOC of this scenario to be 0.68 \$/PAX km. However, this result is viewed with caution, as the estimated DOC appears to be very low and the flight connection under investigation exceeds the current maximum range capabilities of the Lilium Jet.

Overall, the estimated DOCs for the Lilium Jet and the Joby Aviation S4 derived using the DOCEv correspond to the findings of the literature. The estimates obtained for shorter routes are only marginally higher than those reported in the literature, demonstrating an adequate degree of accuracy and completeness. When estimating the DOC for longer regional missions, the results of the DOCEv tend to be reasonably higher

than the estimates of the current literature. This model behaviour is expected and is considered advantageous, as it may be attributed to the superior comprehensiveness of the DOCEv compared to the initial models developed in the literature.

7.2. Delimitation and Extension of the Scientific Literature

The DOC of eVTOL is seldom the primary focus of investigations in the existing literature. Instead, eVTOL DOC is often estimated as part of a broader analysis, such as in network modelling and assessment, or demand forecasting. Examples of this include the research by Goyal et al. (2018), Kohlman et al. (2019), Niklaß et al. (2020), and Ploetner et al. (2020). In these studies, the DOC is integrated into Monte Carlo simulations, system-of-systems approaches, or other modelling techniques. The documentation concerning these estimations is often insufficient and, if documentation is available, it tends to indicate rather abstract approaches for the DOC estimation. For example, Kohlman et al. (2019) estimate the eVTOL maintenance cost based on the assumption that one hour of flight time equates to one hour of maintenance, with an hourly maintenance wrap rate (labour and overhead) of 60 \$/h. Material and battery replacement costs are not mentioned in the documentation.

Only few authors have explicitly focused on estimating eVTOL DOC in their research and usually apply a combination of the general cost estimation approaches. A frequently cited source is Mihara et al. (2021), which estimates the eVTOL DOC for air ambulance operations in Japan. Although their developed model considers several aspects of DOC, it is not suitable for estimating eVTOL DOC from a commercial air transport perspective due to the differences in the operational scenarios. The eVTOL DOC estimation model in the literature that is the closest to the DOCEv was developed by Pertz et al. (2023), who also based their DOC estimation model on the methodology for conventional aircraft provided by Thorbeck (2013). Pertz et al. (2023) also modified this reference methodology to account for eVTOL characteristics and their operation. However, these modifications remain on a more abstract level compared to those in DOCEv.

Despite few similarities in the DOC estimation, the DOCEv offers extensions and advantages compared to the existing literature in different areas of the DOC estimation. For example, the DOCEv encompasses a comprehensive energy cost estimation based on a transparent full performance assessment. The considered eVTOL and flight mission must be specified by the input data, together with the potential contingency, alternate, and/or final reserve mission. Furthermore, the DOCEv goes beyond statistical averages when estimating annual downtimes by incorporating specific operational constraints, such as night curfews and the exceeding of certified weather conditions. Moreover, a detailed turnaround time is projected in the model, considering factors such as boarding processes, charging, and de-icing. In terms of fees, the estimation of handling fees is more detailed due to the integration of aspects such as additional transportation cost for bulky luggage, ground infrastructure utilisation, and eVTOL de-icing. Regarding navigation fees, the DOCEv considers the opportunity of an eVTOL operation under visual flight rules in airspace class E without having ATC contact. Major extensions are also present in the maintenance cost estimation. The developed model is able to consider either in-house maintenance or subcontracted maintenance. In the case of in-house maintenance, existing cost estimation approaches are extended especially with regard to line maintenance, energy system base maintenance, unscheduled maintenance, and overhead cost. In addition to this, the model is also able to differentiate between piloted and autonomous operations, reflecting future operational scenarios.

7.3. Model Limitations

To develop the DOCEv, substantial research has been performed in advance on AAM, eVTOLs and their operation, airline operating costs, and existing DOC estimation methods. However, the DOCEv is subject

to limitations. The DOCEv has been developed primarily to estimate the DOC of an eVTOL. It cannot perform tasks such as identifying passenger demand, locating vertiports, or establishing eVTOL routings. Moreover, the model does not consider the first or last mile, i.e., the way from the passenger's starting point to the departure vertiport and the way from the arrival vertiport to the passenger's final destination. To calculate this, an appropriate network model must also incorporate ways of intermodal transportation.

Further model limitations can be classified into two general categories: data-related and structure-related limitations. Among the data-related limitations is the restricted availability of relevant model input data, as well as the existence of local variations in the data. For instance, input data derived for a case study application in Germany may not be directly applicable to a case study based in the United Arab Emirates, highlighting the challenges of context-specific applicability. In addition to data-related constraints, there are also structure-related limitations inherent to the model. In general, the development of the DOCEv aimed at encompassing as many potential operational circumstances as possible, while maintaining a manageable level of complexity for this initial model version. For example, the model can differentiate between piloted and autonomous eVTOL operations, or between the purchase of the eVTOL energy system and its external provision as a service. However, there are currently no large-scale commercial eVTOL operations. Therefore, the future operational environment, including the eVTOL-related processes at vertiports, during flight, or during maintenance, is subject to uncertainty. Consequently, the overall applicability of the DOCEv may prove constrained in the future if operational scenarios are inaccurately represented. Note that there is a general need for further research in nearly all areas of eVTOL operations, particularly in terms of maintenance and battery degradation. These components are defined based on a limited body of literature in the DOCEv. In addition to these limitations arising from uncertainty, some certain aspects of eVTOL operation are not included in the DOCEv yet, such as demand fluctuations and seasonality, fleet and network effects, leasing of eVTOLs, or the occurrence of empty flights. These limitations currently reduce the applicability of the DOCEv for assessing AAM system designs. However, many of these limitations can be addressed by incorporating additional calculations into the modular structure of the DOCEv, enhancing its capability not only for the DOC estimation but also for the evaluation of AAM system designs.

8. Conclusion and Outlook

8.1. Summary of the Conducted Work

eVTOLs are considered to have great potential to complement or substitute existing ground transportation. To evaluate the economic feasibility of eVTOL operations in terms of DOC, the DOCEv has been developed as a novel estimation model. It integrates a full eVTOL performance analysis and an eVTOL utilisation estimation, factoring in annual downtime, turnaround time, and block time. Furthermore, the DOCEv estimates the cost associated with ownership, insurance, crew, energy, fees, and maintenance, and provides an approach for indicative ticket pricing.

The DOCEv is implemented in Microsoft Excel and is provided as a tool, allowing flexible and transparent DOC estimation. Its modular architecture considers each DOC component separately, allowing users to omit or modify specific components, as well as incorporate additional factors as needed. Consequently, most structure-related limitations can be addressed by users themselves, leaving input data availability as a primary constraint. If the necessary input data is available, the model can be applied across a wide range of scenarios, including urban and regional operations, as well as European and global contexts. However, transferring DOC estimates between scenarios should be approached with caution, as results may vary significantly depending on the operational and geographic context.

The applicability of the DOCEv has been demonstrated in a case study for an airport shuttle between MUC and SZG. It became apparent that the DOC estimates are marginally higher for shorter, regional routes of approximately 80 km great-circle distance compared to values reported in the literature. For longer routes, the DOCEv estimates are reasonably higher than those found in existing studies. Against this background, the DOCEv is considered as a comprehensive and accurate model for the detailed estimation of eVTOL DOC.

In summary, the DOCEv has significant potential for application in the ongoing exploration of eVTOL DOC. Its modular design allows for easy modifications and extensions, making the model highly robust and well-suited for application within the dynamic environment of AAM.

8.2. Future Research Needs

Future work related to the DOCEv should involve the validation of the provided input data set and the eVTOL operations represented in the model. Furthermore, the influence of potential empty flights during on-demand operations on the generated ticket prices and different ticket fare structures can be investigated. The modular structure of the DOCEv allows for further modification and extension. An example for this could be the integration of IFR operations, eVTOL leasing, or fleet and network effects. In the future, the DOCEv can be applied to further investigate eVTOL DOC by performing sensitivity analyses.

Finally, eVTOLs are expected to contribute to a more sustainable transport sector. Against this background, life cycle assessment should be integrated into the DOCEv to determine environmental impacts besides the DOC and investigate potential trade-offs. In the future, the DOCEv may be extended and integrated into a collaborative approach for the development, modelling, and assessment of entire AAM concepts.

CRedit authorship contribution statement

Felix Müller: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alexander Barke:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Olof Nittinger:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Thomas S. Spengler:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jairtraman.2025.102951>.

Data availability

The data applied is provided in the Supporting Information S1.

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